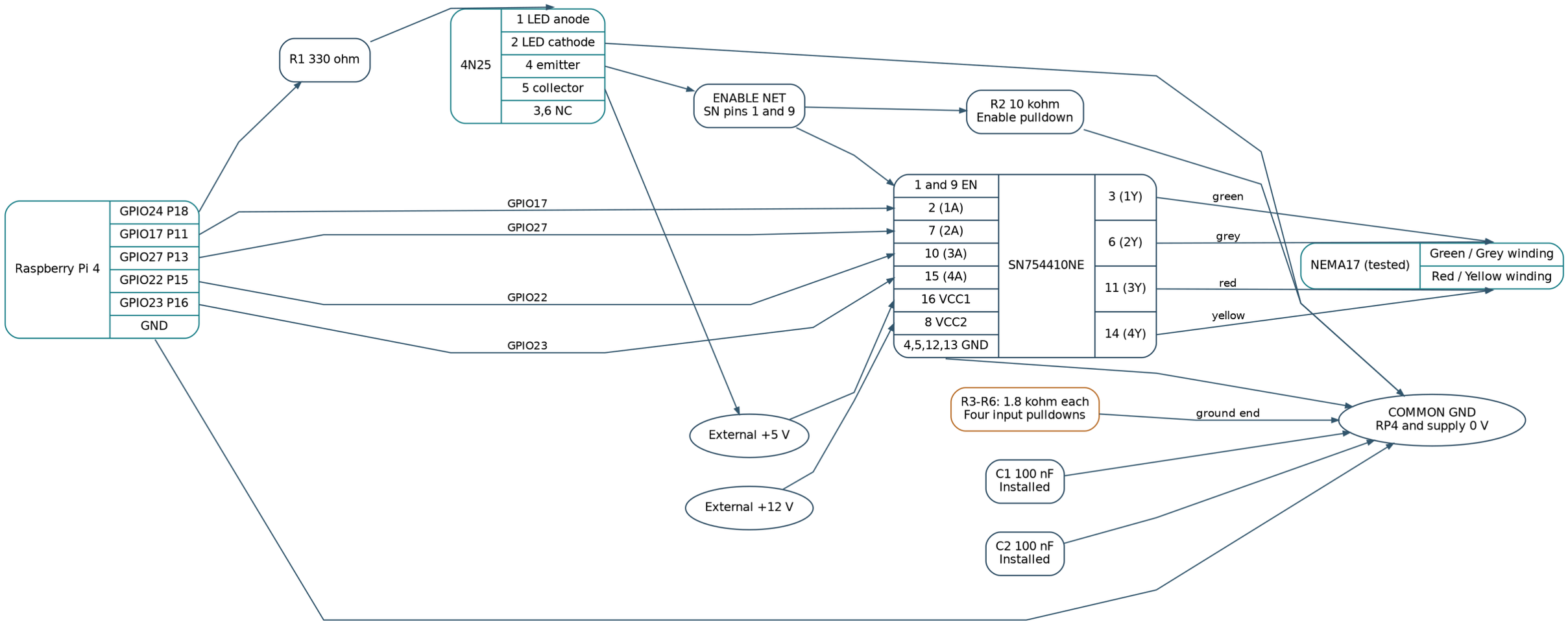
PARMCO - CHECKPOINT 1 | COMPONENT CONNECTIONS

Documented wiring with installed 1.8 kohm input pulldowns and 100 nF bypass capacitors



R3-R6: one 1.8 kohm resistor from EACH SN input (pins 2, 7, 10, 15) to common ground.

C1: 100 nF VCC1 (pin 16) to GND; C2: 100 nF VCC2 (pin 8) to GND. Both reported installed.

Note: resistor/capacitor branches are specified here and on the following pin-level schedule; diagram groups parallel branches.

Physical direction not independently confirmed; clockwise and counterclockwise control patterns were measured and verified.

PIN-LEVEL NETLIST | FINAL REPORTED VALUES

From	Component / connection	To
RP4 GPIO24 (physical pin 18)	R1: 330 ohm series	4N25 pin 1 (LED anode)
4N25 pin 2	Direct wire	RP4/common GND
External +5 V	Direct wire	4N25 pin 5 (collector)
4N25 pin 4 (emitter)	Enable node	SN754410NE pins 1 and 9
Enable node	R2: 10 kohm pulldown	Common GND
RP4 GPIO17 (physical pin 11)	Direct wire	SN pin 2 (1A)
RP4 GPIO27 (physical pin 13)	Direct wire	SN pin 7 (2A)
RP4 GPIO22 (physical pin 15)	Direct wire	SN pin 10 (3A)
RP4 GPIO23 (physical pin 16)	Direct wire	SN pin 15 (4A)
SN pin 2 (1A)	R3: 1.8 kohm pulldown	Common GND
SN pin 7 (2A)	R4: 1.8 kohm pulldown	Common GND
SN pin 10 (3A)	R5: 1.8 kohm pulldown	Common GND
SN pin 15 (4A)	R6: 1.8 kohm pulldown	Common GND
External +5 V	Direct wire	SN pin 16 VCC1
External +12 V	Direct wire	SN pin 8 VCC2
SN pins 4, 5, 12, 13	Direct wires	Common GND (also RP4 and supply GND)
SN pin 16 VCC1	C1: 100 nF bypass, installed	Common GND
SN pin 8 VCC2	C2: 100 nF bypass, installed	Common GND
SN pins 3 and 6	Motor winding (~37 ohm)	Green and grey leads
SN pins 11 and 14	Motor winding (~36.3 ohm)	Red and yellow leads

Verified electrically: four GPIO inputs, clockwise and counterclockwise step patterns, 4N25 enable HIGH, timer frequency 54 MHz.

Motor test: lab partner reported successful operation; exact physical direction, one-revolution accuracy and stall-speed data pending.

Stand installed. Do not rewire powered equipment. SN754410NE has no automatic winding-current regulation.